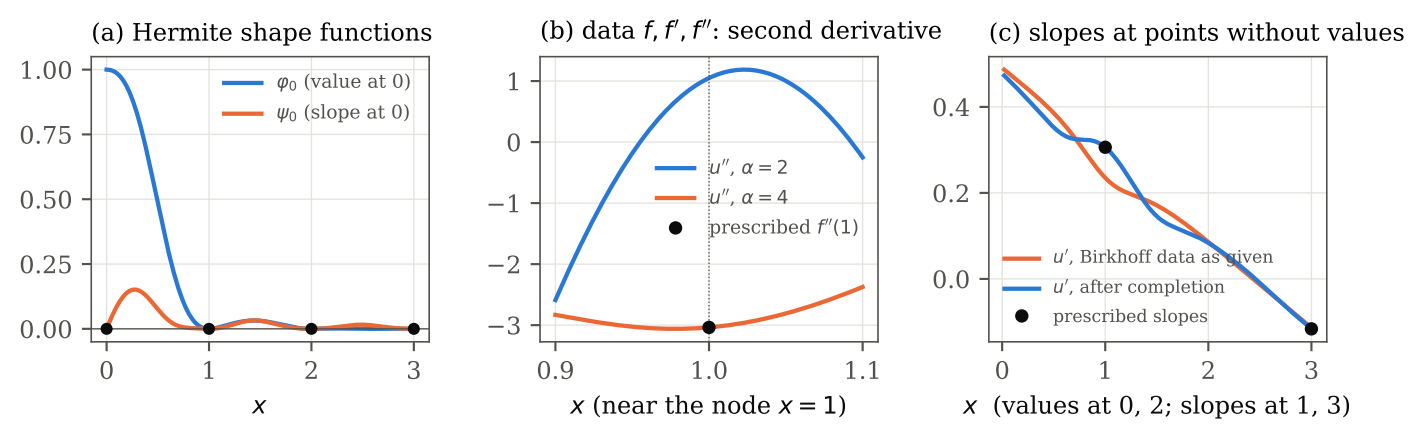
(a) Hermite shape functions
1.00
0.75
0.50
0.25
0.00
0
1
2
3
x
φ0 (value at 0)
ψ0 (slope at 0)
(b) data f, f′, f″: second derivative
1
0
−1
−2
−3
0.9
1.0
1.1
x (near the node x = 1)
u″, α = 2
u″, α = 4
prescribed f″(1)
(c) slopes at points without values
0.4
0.2
0.0
0
1
2
3
x (values at 0, 2; slopes at 1, 3)
u′, Birkhoff data as given
u′, after completion
prescribed slopes